

A Practical Guide to Reinforcement learning

TOPIC -- REINFORCEMENT LEARNING

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$R_a(s, s')$, the immediate reward after transitioning from s to s' under action a . The purpose of reinforcement learning is for the agent to learn an optimal (or near-optimal) policy that maximizes the reward function or other user-provided reinforcement signal that accumulates from immediate rewards. This is similar to processes that appear to occur in animal psychology. For example, biological brains are hardwired to interpret signals such as pain and hunger as negative reinforcements, and interpret pleasure and food intake as positive reinforcements. In some circumstances, animals learn to adopt behaviors that optimize these rewards. This suggests that animals are capable of reinforcement learning. A basic reinforcement learning agent interacts with its environment in discrete time steps. At each time step t , the agent receives the current state S_t and reward R_t . It then chooses an action A_t from the set of available actions, which is subsequently sent to the environment. The environment moves to a new state S_{t+1} and the reward R_{t+1} associated with the transition (S_t, A_t, S_{t+1}) is determined. The goal of a reinforcement learning agent is to learn a policy:

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A set of environment and agent states (the state space), S ; A set of actions (the action space), A , of the agent;

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In situation s perform action a . Receive a consequence situation s' . Compute state evaluation $v(s')$ of how good is to be in the consequence situation s' . Update crossbar memory $w'(a, s) = w(a, s) + v(s')$. Initial conditions of the memory are received as input from the genetic environment. It is a system with only one input (situation), and only one output (action, or behavior). Self-reinforcement (self-learning) was introduced in 1982 along with a neural network capable of

self-reinforcement learning, named Crossbar Adaptive Array (CAA). The CAA computes, in a crossbar fashion, both decisions about actions and emotions (feelings) about consequence states. The system is driven by the interaction between cognition and emotion.

-- 04 --

The algorithms then adjust the weights, instead of adjusting the values associated with the individual state-action pairs. Methods based on ideas from nonparametric statistics (which can be seen to construct their own features) have been explored. Value iteration can also be used as a starting point, giving rise to the Q-learning algorithm and its many variants. Including Deep Q-learning methods when a neural network is used to represent Q, with various applications in stochastic search problems. The problem with using action-values is that they may need highly precise estimates of the competing action values that can be hard to obtain when the returns are noisy, though this problem is mitigated to some extent by temporal difference methods. Using the so-called compatible function approximation method compromises generality and efficiency.

-- 05 --

Associative reinforcement learning Associative reinforcement learning tasks combine facets of stochastic learning automata tasks and supervised learning pattern classification tasks. In associative reinforcement learning tasks, the learning system interacts in a closed loop with its environment.

-- 06 --

Self-reinforcement learning Self-reinforcement learning (or self-learning), is a learning paradigm which does not use the concept of immediate reward $R_a(s, s')$ after transition from s to s' with action a . It does not use an external reinforcement, it only uses the agent internal self-reinforcement. The internal self-reinforcement is provided by mechanism of feelings and emotions. In the learning process emotions are backpropagated by a mechanism of secondary reinforcement. The learning equation does not include the immediate reward, it only includes the state evaluation. The self-reinforcement algorithm updates a memory matrix $W = [w(a, s)]$ such that in each iteration executes the following machine learning routine:

-- 07 --

In machine learning and optimal control, reinforcement learning (RL) is concerned with how an intelligent agent should take actions in a dynamic

environment in order to maximize a reward signal. Reinforcement learning is one of the three basic machine learning paradigms, alongside supervised learning and unsupervised learning. While supervised learning and unsupervised learning algorithms respectively attempt to discover patterns in labeled and unlabeled data, reinforcement learning involves training an agent through interactions with its environment. To learn to maximize rewards from these interactions, the agent makes decisions between trying new actions to learn more about the environment (exploration), or using current knowledge of the environment to take the best action (exploitation). The search for the optimal balance between these two strategies is known as the exploration–exploitation dilemma.

-- 08 --

Function approximation methods In order to address the fifth issue, function approximation methods are used. Linear function approximation starts with a mapping ϕ that assigns a finite-dimensional vector to each state-action pair. Then, the action values of a state-action pair (s, a) are obtained by linearly combining the components of $\phi(s, a)$ with some weights θ :

-- 09 --

Generalization and transferability The RL agents trained in specific environments often struggle to generalize their learned policies to new, unseen scenarios. This is the major setback preventing the application of RL to dynamic real-world environments where adaptability is crucial. The challenge is to develop such algorithms that can transfer knowledge across tasks and environments without extensive retraining.

-- 10 --

actor-critic architecture actor-critic-scenery architecture adaptive methods that work with fewer (or no) parameters under a large number of conditions bug detection in software projects continuous learning combinations with logic-based frameworks (e.g., temporal-logic specifications, reward machines, and probabilistic argumentation). exploration in large Markov decision processes entity-based reinforcement learning human feedback interaction between implicit and explicit learning in skill acquisition intrinsic motivation which differentiates information-seeking, curiosity-type behaviours from task-dependent goal-directed behaviours large-scale empirical evaluations large (or continuous) action spaces modular and hierarchical reinforcement learning multiagent/distributed reinforcement learning is a topic of interest. Applications are expanding. occupant-centric control optimization of computing resources

partial information (e.g., using predictive state representation) reward function based on maximising novel information sample-based planning (e.g., based on Monte Carlo tree search). securities trading transfer learning TD learning modeling dopamine-based learning in the brain. Dopaminergic projections from the substantia nigra to the basal ganglia function are the prediction error. value-function and policy search methods